EVALUATION OF HIGHWAY BARRIER PERFORMANCE WITH ETC LANES

Computer Engineering

Project work of
Performance Evaluation of Computer Systems and Networks

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INTRODUCTION

PROBLEM DESCRIPTION

- Highway toll barrier with N lanes, K of which dedicated to vehicles with ETC device.
- Only a percentage p of the total number of vehicles is equipped with the ETC device.
- The service time of a truck is 1.5 times the one of a car.
- The service time of a ETC lane is costant.
- Vehicles select the lane with the smallest queue among those they are permitted to use.

OBJECTIVE: Studying the behavior of the whole highway toll barrier with different combinations of K and p.

PERFORMANCE INDEXES:

- E[R]: mean response time
- E[W]: mean waiting time

MODELING

Some **general assumptions** have been made and analysed in the part of **preliminary validation**; the most important are:

- The vehicle of the highway coincides with the job of the system.
- A vehicle is assigned to the lane with the least number of vehicles in the queue and it cannot change its lane.
- In case of a tie between two or more lanes, the lane assigned will be the one with the minimum id.
- There are not hybrid lanes.

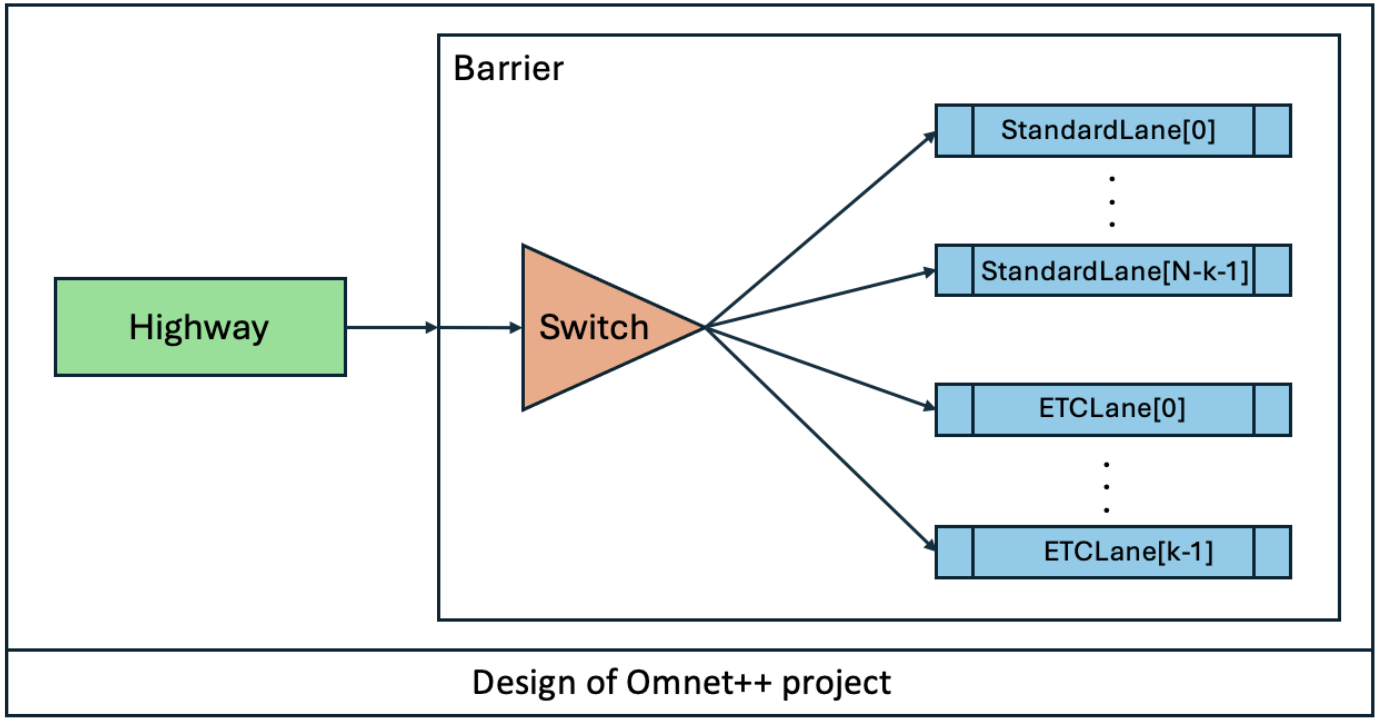
The **factors** that may affect the performance of the system are:

- **N**, total lanes of the Highway toll barrier.
- **K**, lanes dedicated to vehicles equipped with ETC device.
- **p**, percentage of vehicles equipped with ETC device.
- $\mathbf{T_c}$, mean interarrival time for cars.
- $\mathbf{T_t}$, mean interarrival time for trucks.
- **S**, mean service time of “Standard lanes”.
- **S_etc**, service time of “ETC lanes”.
- **stddev**, standard deviation of the lognormal distribution

IMPLEMENTATION

We have defined a new format of messages to properly represent cars and trucks, called *Vehicle*.
Its fields are:

- bool **isCar;**
- bool **hasEtc**;
- simtime_t **arrivalTime**.



VERIFICATION 1/2

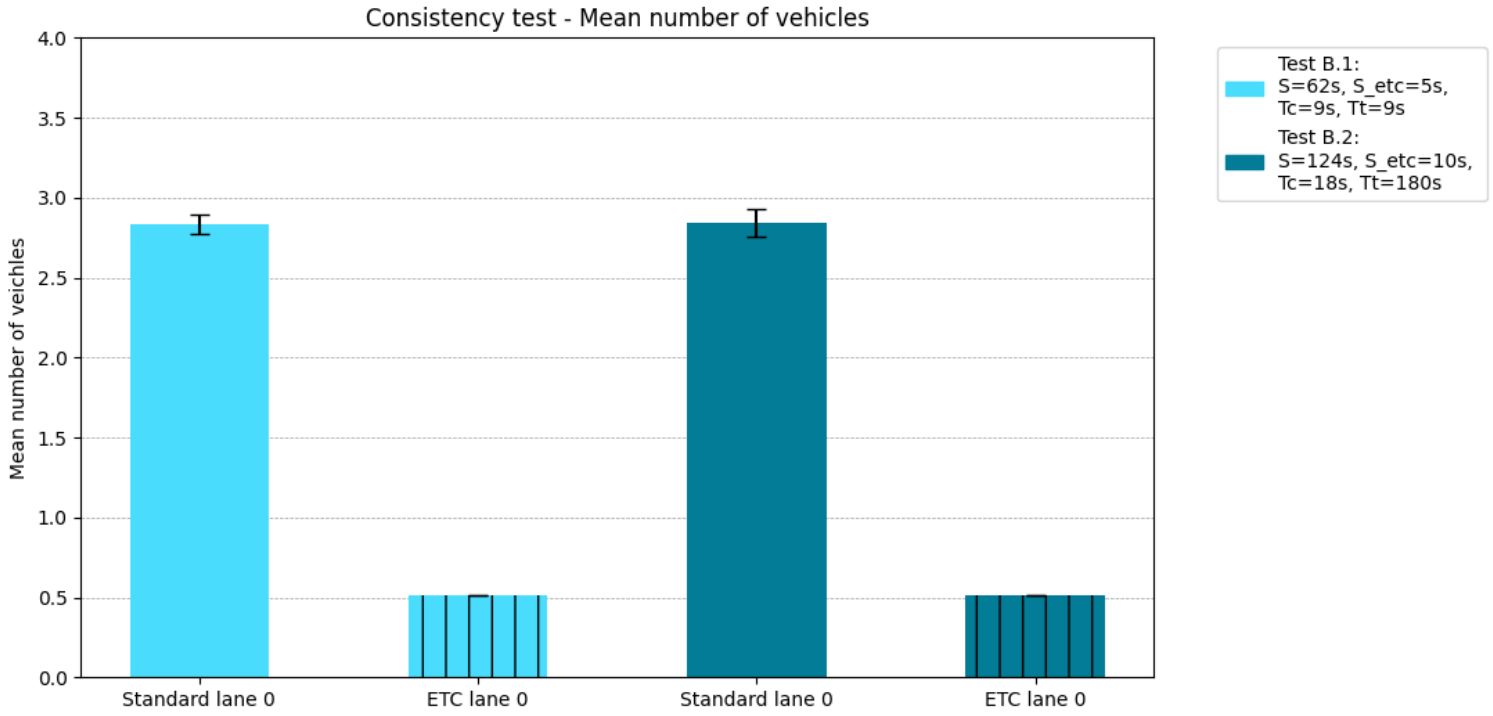
DEGENERACY TEST

Analysis of model behavior under degenerate conditions, setting parameters to 0 or extreme values:

- First test: K=0, no ETC lanes;
- Second test: p=0 and 0<K<N, no vehicles equipped with ETC device.

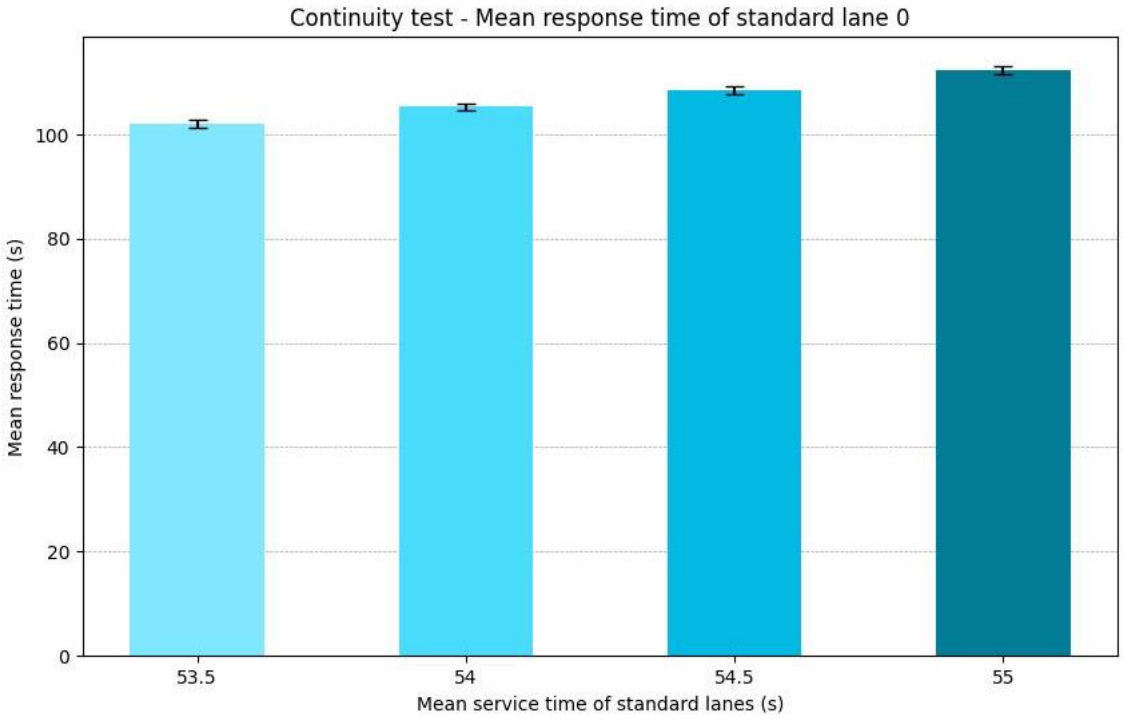
CONSISTENCY TEST

To analyse the relationship between the arrival rate of vehicles and the service time of lanes. In this case, we doubled both the interarrival time of vehicles and the service time of both types of lanes.

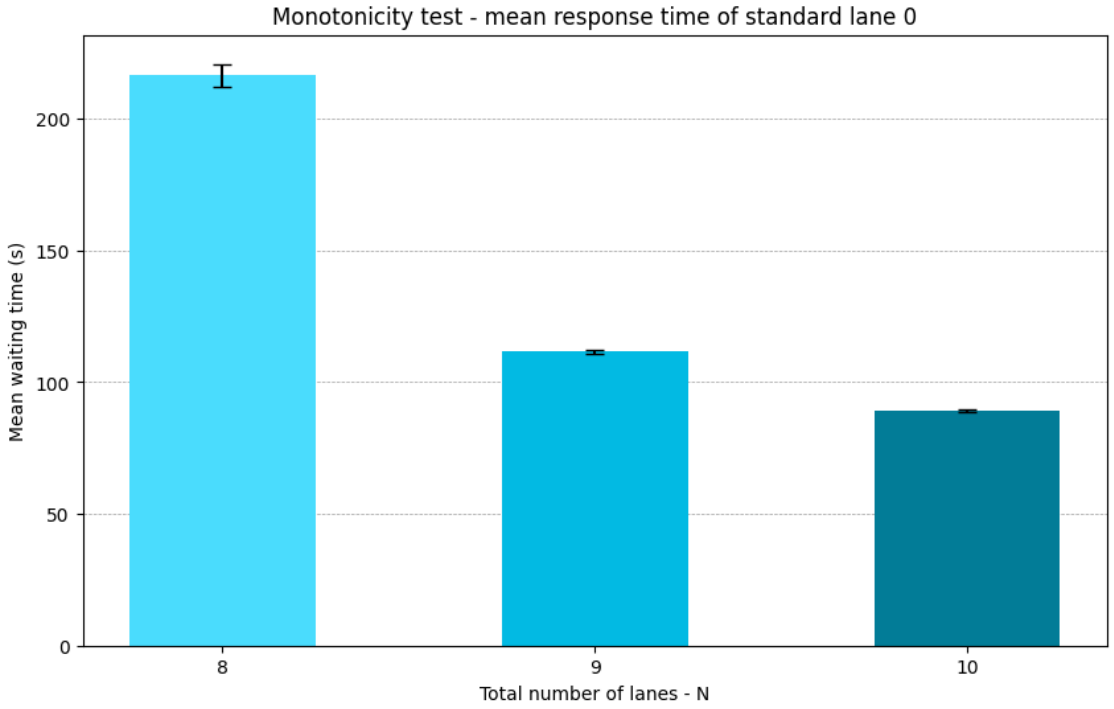


VERIFICATION 2/2

CONTINUITY TEST

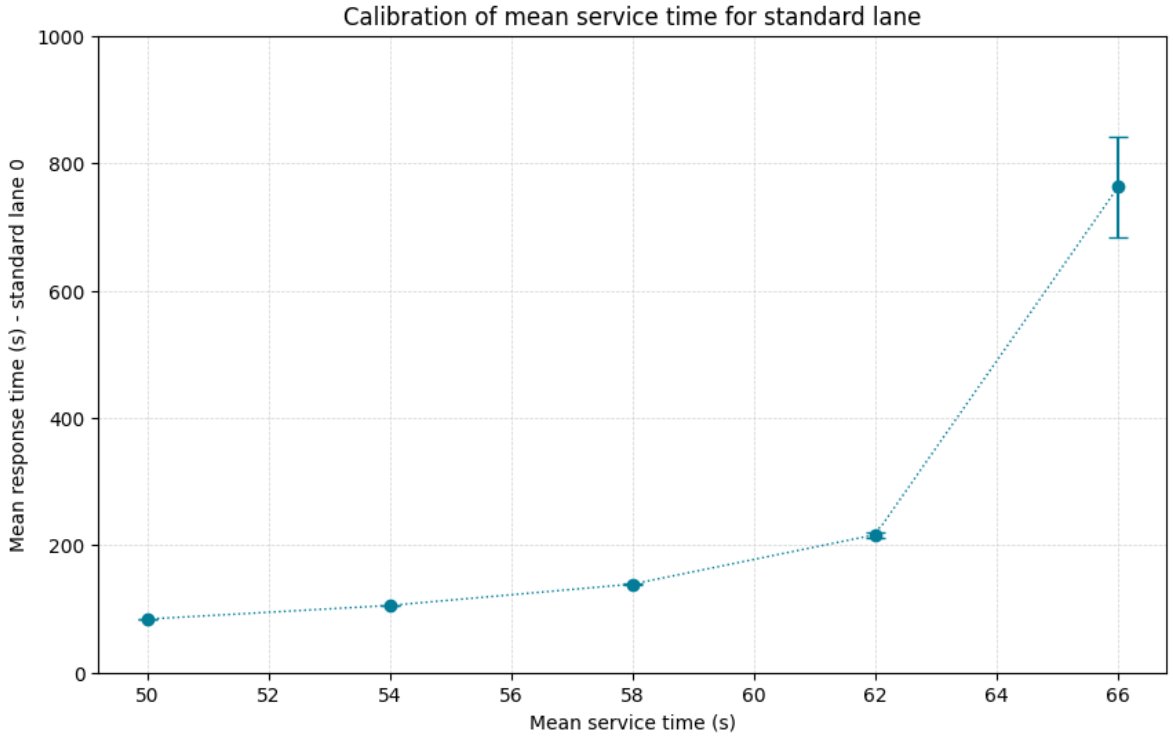


MONOTONICITY TEST

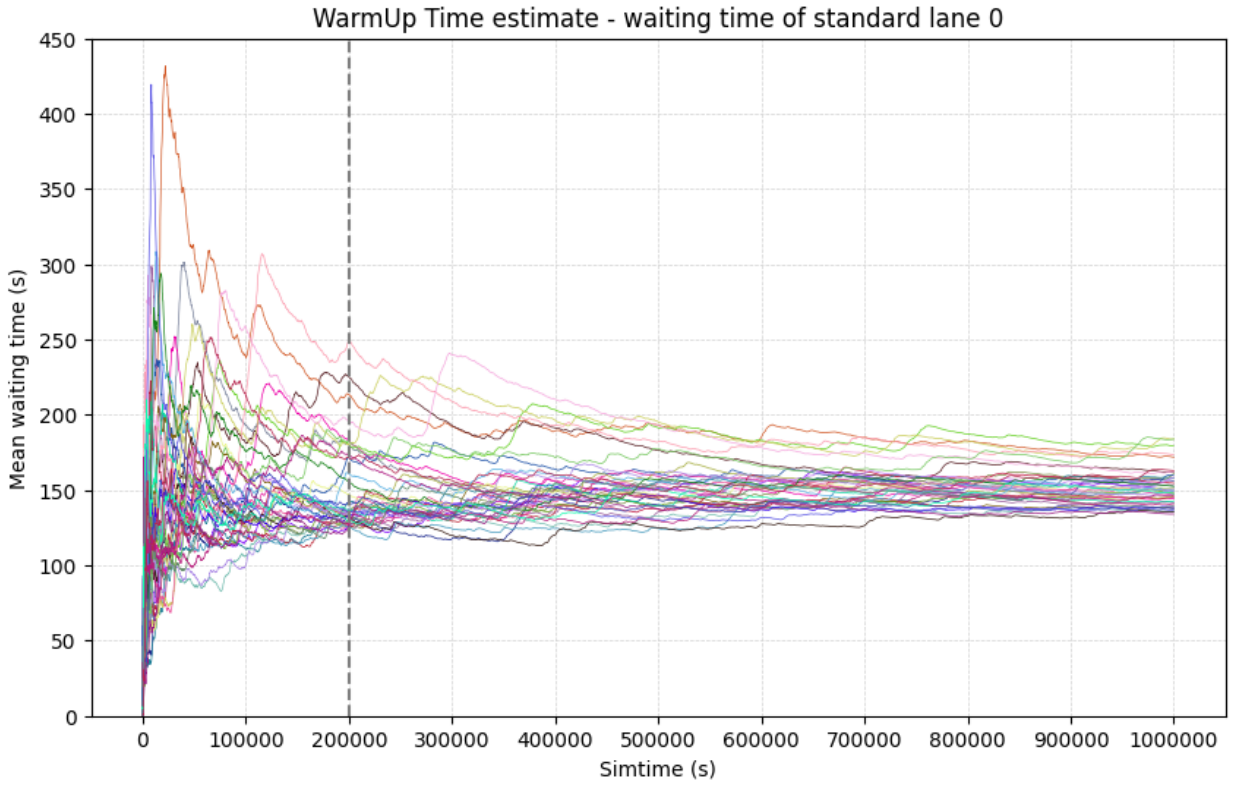


CALIBRATION

MEAN SERVICE TIME



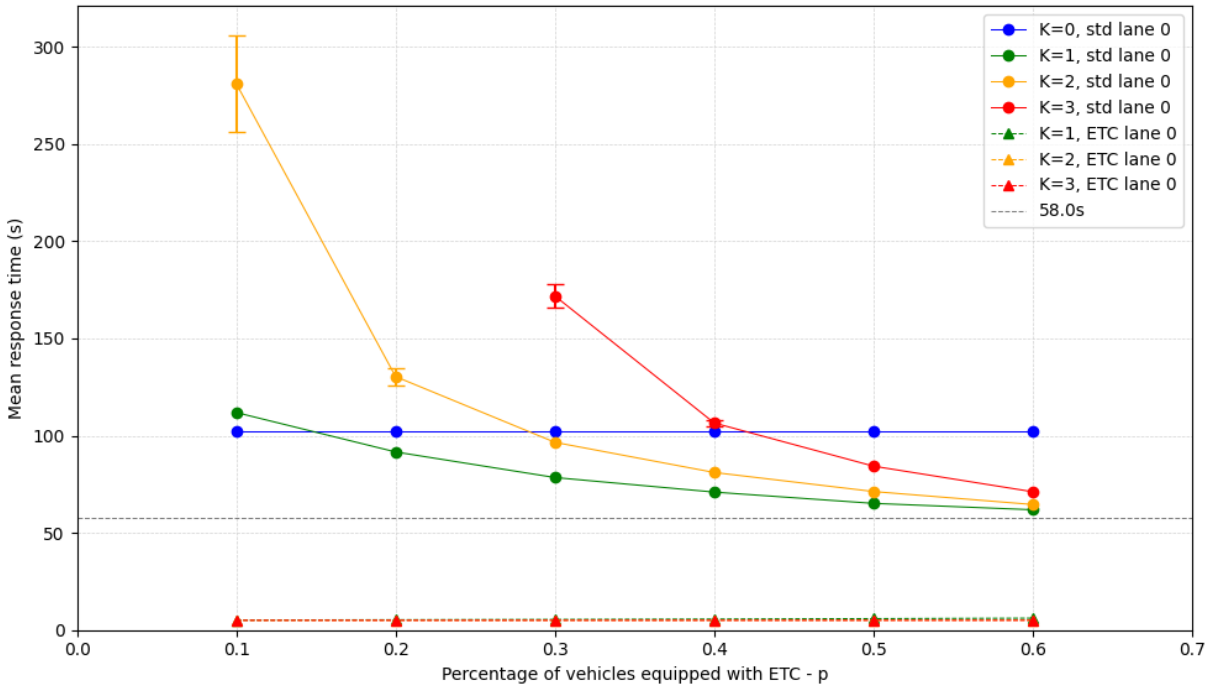
WARM-UP TIME



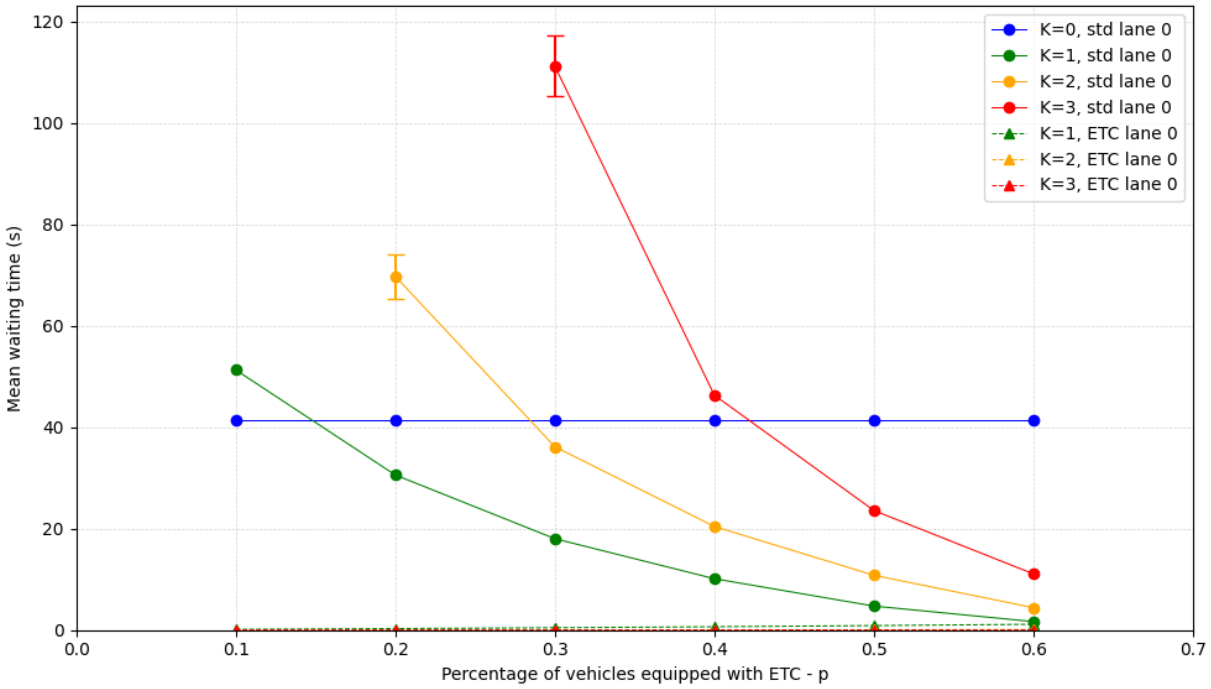
SIMULATION EXPERIMENTS 1/2

EXPONENTIAL DISTRIBUTION – MEDIUM WORKLOAD

RESPONSE TIME



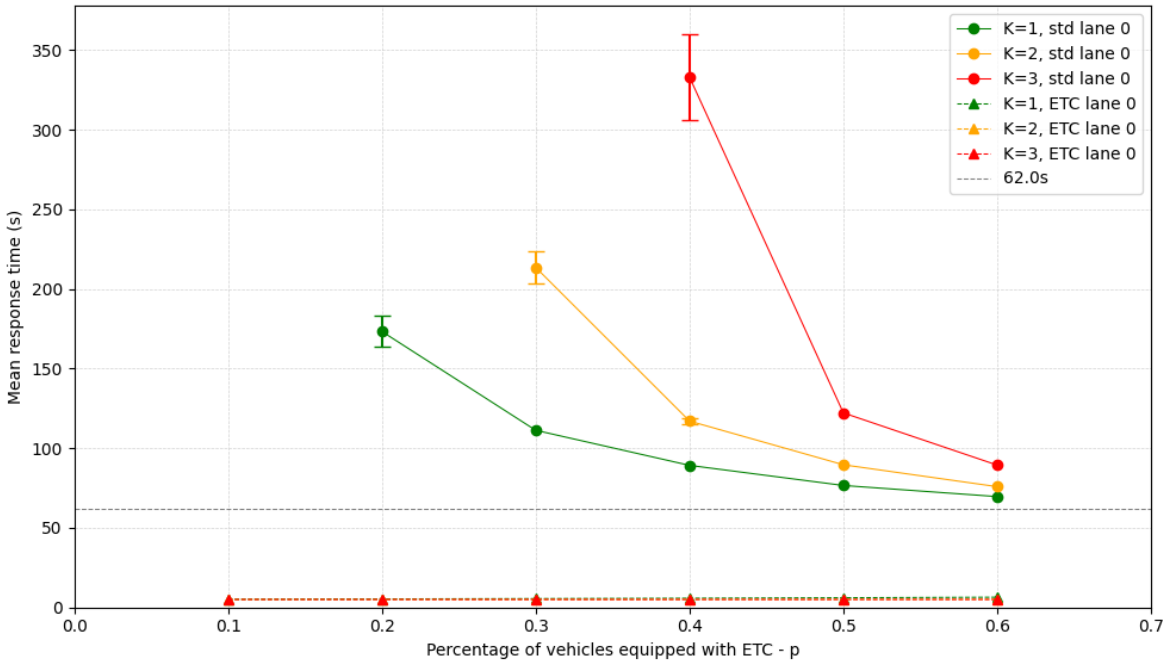
WAITING TIME



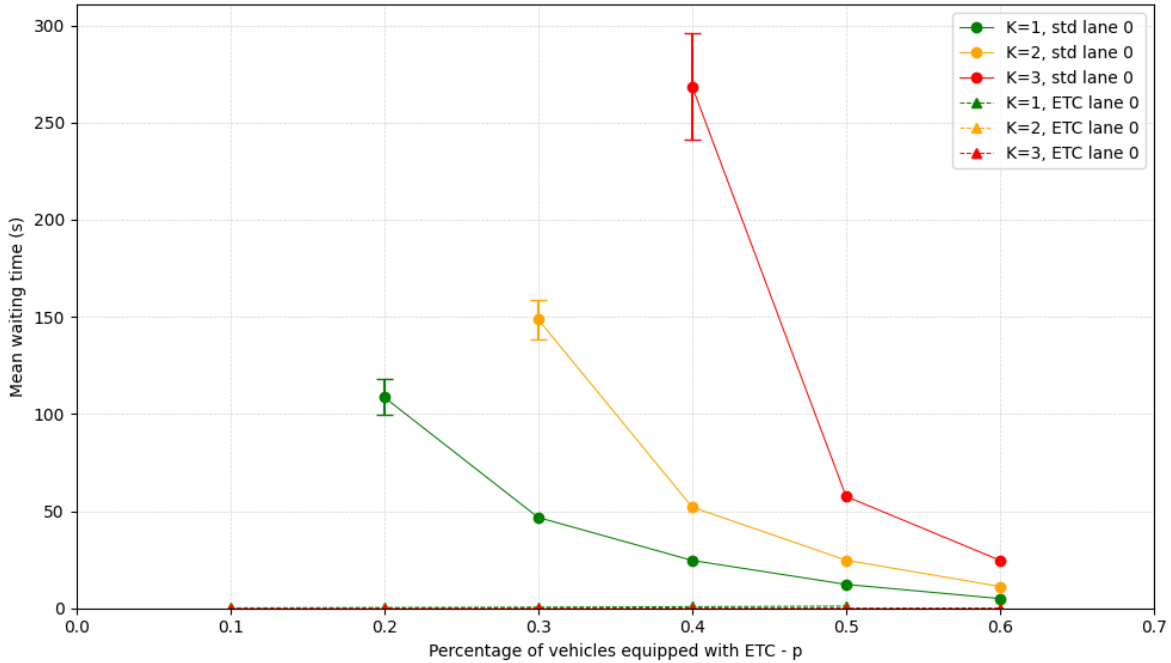
SIMULATION EXPERIMENTS 2/2

EXPONENTIAL DISTRIBUTION – HIGH WORKLOAD

RESPONSE TIME



WAITING TIME



CONCLUSION

The **presence of ETC lanes** is fundamental not only for significantly reducing the waiting times for ETC-equipped vehicles but also for improving the overall performance of the system.

Keeping constant the number of total lanes N, the increase of K (number of ETC lanes) could be a risk because **N-K standard lanes could be not enough** to handle the vehicles incoming, leading to congestion and increasing the waiting times in those standard lanes.

We have a significant improvement even in standard lanes only where the percentage p is high, so that a significant part of vehicle flow is assigned to the ETC lane(s).

These observations apply to all configurations we have tested. Higher workloads increase demand for standard lanes, and an increasing standard deviation in the lognormal distribution increases waiting and response times without changing the overall system behavior.